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New Generation of Expandable Diverter for Well Stimulation: Preparation, Properties and Performance Evaluation

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Abstract

To enhance the productivity of unconventional reservoirs, it is essential to create extensive fractures or worm-like networks in the formation during well stimulation. Using effective diverters can increase the number of fractures, thereby reduce the number of fracturing stages and consequently lower operational costs and boost well productivity.

To address the challenge, a new generation of diverter has been developed using elastic polymeric particles with good expandability and degradability, based on polyurethane chemistry. The effect of synthesis conditions, such as the prepolymer concentration and agitation speed on the particle size and material properties has been studied. The chemical composition, thermal stability, morphology/microstructure, and elasticity of these materials have been characterized using FTIR, TGA, SEM and uniaxial compression tests. The plugging performance of the polymeric diverter was examined.

By adjusting the synthesis conditions, the particle size of the synthesized polymeric diverter can be easily tuned in the range of 0.2 – 8 mm. Characterization results confirm the presence of both polyurea and polyurethane structures, as well as good thermal stability for subsurface wellbore operations. Compression experiments demonstrate the excellent elasticity of the materials, showing that the diverter can quickly return to its original shape and size once the compression stress is released. In the plugging performance tests, the diverter material completely blocked fluid flow at a slot size of 2 mm and 4 mm. As the slot size gradually increased from 0.3 mm to 3 mm, no drop in differential pressure was observed, indicating continuous sealing during multiple fracture initiation and propagation processes. The dynamic temporary plugging performance evaluation clearly shows the efficiency of the newly developed diverter, thus enhancing well productivity.

The newly developed diverter material shows great promise for diversion applications. It can improve the efficiency of multi-staged fracturing, thereby reducing the total number of stages, and ultimately the associated operational time and costs. Furthermore, due to its unique properties and degradability, this material holds high potential for use in water shut-off application as well.

Introduction

Due to the heterogeneity and unfriendly physical properties of reservoirs like low permeability, well stimulation is often carried out to boost the production of hydrocarbons from the wellbore, especially in the unconventional oil and gas reservoirs. Among various well stimulation treatments, hydraulic fracturing and acidizing are the two most common approaches. As the name implies, hydraulic fracturing is to fracture the subterranean formation by injecting a fracturing fluid such as water, supercritical carbon dioxide, at a high pressure and holding up the created fractures by the proppant carried subsequently, thereby increasing the oil and gas inflow capacity. Acidizing involves injecting strong acid such as hydrofluoric acid, hydrochloric acid, or their combinations into the subterranean formation to dissolve the rock and create macroscopic channels (or wormholes), which makes the oil and gas flow faster and easier, consequently enhances the well productivity. However, the conventional fracturing and acidizing is less effective when the zone to be stimulated is long and heterogeneous with large inter-formation permeability coefficients of variation and the natural fractures as the fracturing or acidizing fluid will naturally flow to the area with lower resistance or higher permeability.(Elsharafi and Bai 2016, Wang, Liu et al. 2022)

To overcome the disadvantage, diversion technology is usually applied along with the hydraulic fracturing and acidizing, which is a commonly used method to increase the stimulated reservoir volume (Chang, Qiu et al. 2007, Vernáez, García et al. 2016, Van Domelen 2017, Chen, Wang et al. 2025). It uses a material so-called diverter to temporarily plug the high permeable areas, such as natural fracture, previously created fractures or high permeability layers, and redirects the fracturing/acidizing fluid to the areas with low permeability to create new fractures or wormholes and achieve more homogeneous stimulation(Han, Geng et al. 2020, Wang, Huang et al. 2020, Wang, Liu et al. 2022).

Diverter can be classified into two categories: mechanical and chemical diverters (Li and Sahu 2023, Bist, Nair et al. 2024). Mechanical diverters, such as ball sealers, can effectively control the fluid entry point, but their use is limited by high cost and risk associated with mechanical intervention into a wellbore. (Bist, Nair et al. 2024) Chemical diverters, such as particulates, are also often applied to divert the flow of fracturing/acidizing fluid, which is achieved by physically blocking the fluid flow through particle bridging with the particulates. In this case, the particulates should seal the fractures or wormholes (openings) tightly with different sizes and/or shapes, maintain the sealing throughout the entire stimulation process and finally degrade fully after treatment to avoid any potential damage to the formation.

Traditional particulate diverters such as PLA are normally rigid with fixed size/shape, while the openings of fractures and wormholes may vary in terms of size and shape. Even within the same well, the openings may have a broad size distribution. Consequently, particulates with the same shape, size and particle size distribution may not be able to effectively and efficiently serve its purpose. To achieve desired behavior, it often requires high dosage. In addition, the dimension of the openings may change during the stimulation process. For example, the width of the sealed opening may increase due to the additional pressure for the 2nd hydraulic fracturing or continuous dissolution of rock by acid during the acid treatment. In that case, the particulate-based diverters with fixed sizes may not be able to continue sealing the openings tightly. As a result, the fracturing fluid will be reversely diverted back into the existing fracture instead of being used to propagate the new fracture. Thus, inconsistent PLA diverter performance, wide variation in recommendations and effectiveness have been frequently reported (Trumble, Sinkey et al. 2019, Li and Sahu 2023).

One solution is to blend deformable particles into the particulates with fixed size/shape (Dreyer, Mazon et al. 2024). The incorporation of deformable particles can make the diverter more tolerant to a wider opening/fracture size distribution, because deformable particles can swell to various sizes to maintain reasonable sealing efficiency. But it is not able to resolve the issue induced by the change in dimensions of the fracture/wormhole opening during the stimulation process.

In this paper, we report a newly developed compressible/elastic/expandable diverter material to address the challenge. The new diverter can be squeezed into the fracturing or wormhole openings under pressure (or compression) and form particles park sealing the openings. When the dimension of the openings widens, the mechanically elastic diverter particles expand automatically and continuously push against the walls of the openings to maintain sealing efficiency. Here, highly elastic diverter particles with size ranging from 0.2 to several millimeters (mm) can be obtained with this new formulation and preparation method. The effect of synthesis conditions, such as the prepolymer concentration and agitation speed on the particle size and material properties has been studied. The chemical composition, thermal stability, morphology/microstructure, and elasticity of these materials have been characterized using FTIR, TGA, electronic microscope and uniaxial compression tests. The dynamic temporary plugging performance of the polymeric diverter was systematically examined.

Experimental materials and methods

Diverter preparation

The synthesis of the new expandable elastic diverter is based on polymer chemistry, which is conducted under mild reaction conditions and the process is easy to scale up. The topology of polymer particles can be controlled and tuned by the agitation energy, reaction conditions, starting formulations, size of pipette, and post-synthesis treatment. The synthesized polymer particles are collected and bottled for further analysis and performance evaluation.

Diverter characterization

A VHX digital microscope (Keyence Corp.) was used to examine the size and shape of the synthesized diverter particles.

A Nicolet iS10 FT-IR spectrometer (Thermo Scientific) was used to obtain the FTIR spectra of the prepolymer and the diverter samples for their chemical structure. The instrument was operated in the transmission mode and collected at room temperature. KBr was used as the background. The IR resolution was 4 cm⁻¹.

The thermostability of the prepared diverter was examined by thermogravimetric analysis method, which is conducted on the SDT Q-600 TGA (TA Instruments Inc.). About 10 mg of sample was weighed and loaded to the sample pan for TGA measurement. The temperature increased from room temperature to 700 °C under inert nitrogen at the flow rate of 100 mL/min. The temperature ramp rate was 10 °C/min. The weight loss versus temperature curve of the diverter sample was obtained.

A universal testing machine (INSTRON 5966, Instron, USA) was used to test the mechanical performance of the synthesized diverter particles. The test consisted of 4 cycles of uniaxial compression and subsequent release of the compression in the same direction. The compression stress was fixed at 2000 psi. During the test, the Bluehill 3.0 software recorded the compression stress and change in the compression strain, generating the stress-strain curve. The rate of compression was set at 60 mm/min. The compression tests were conducted at room temperature (25 ± 2 °C). The precision in tracking the displacement is ±0.01 mm, and the resolution for the load measurement is ±0.05N.

The energy loss coefficient ($\Delta U/U$) that used to estimate the elasticity of samples could be calculated based on the stress-strain curve using the following equation (Qiu, Zhu et al. 2024):

$$\frac{\Delta U}{U} = \frac{E_{\text{loading}} - E_{\text{unloading}}}{E_{\text{loading}}} \times 100\%,$$

where E_{loading} is the integral area of the stress-strain curve in the loading phase, and $E_{\text{unloading}}$ is the integral area of the stress-strain curve in the unloading phase.

The Young's modulus (E_Y) of the diverter sample is estimated by the slope of the loading curve at the beginning stage. The compressive stress (σ) is defined as the stress at a strain of 10% (Qiu, Zhu et al. 2024).

Evaluation of diverter's plugging performance

High pressure slot testing was performed using the apparatus shown in Figure 1. It comprises a 1-in. pressure graded stainless-steel tube ending with a slot insert. A variety of output slots with widths were used to mimic fractures of various widths.

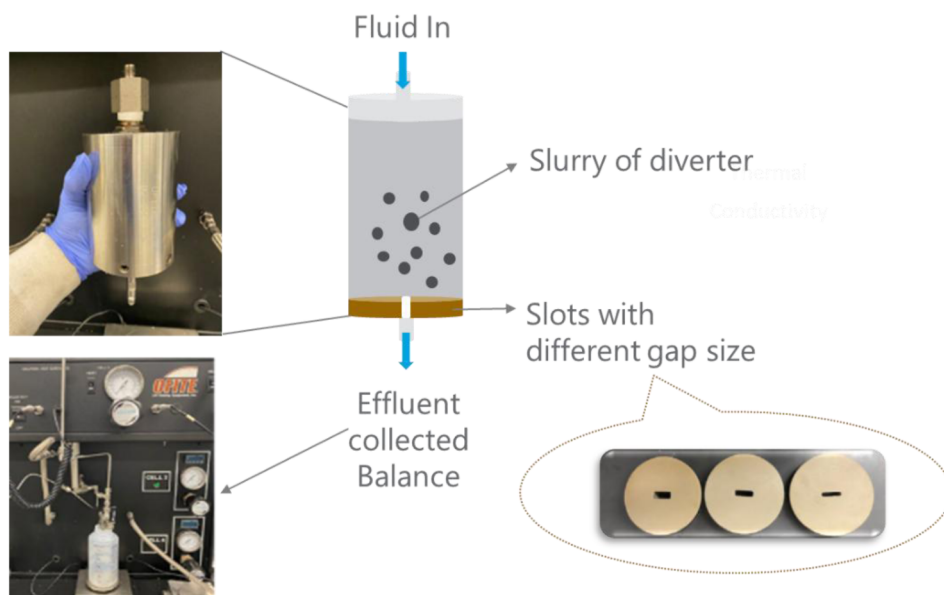


Figure 1—Experimental setup used to check the bridging and plugging performance of diverters.

During the test, about 2 ppg of the diverter particles were added to a 20-lbm/1000-gal linear gel prepared with guar. The gel was crosslinked with a borate crosslinker. The slurry was poured into and filled the tube to fill the entire. Another batch of borate crosslinked guar gel was prepared and filled in another container as the fracturing fluid. A plug was installed behind the slot to avoid leaking of the slurry. A pressure gauge was used to measure the diverting pressure throughout the test. When starting, the plug is then removed and the fracturing fluid was pumped from its container at a constant pumping pressure (1000 psi, showing a variable flow rate varied with pressure difference). Under pumping, the slurry was then squeezed through the slot. At bridging, a plug forms, which caused accumulation of material and eventually a drastic increase in pressure drop, up to 700 psi, at which point test was stopped. The threshold of 700 psi was chosen to represent a reasonable overlap between the real pressure drops observed in the field and the laboratory pump/pipe operating limits.

To obtain plugging performance under dynamic conditions, another testing device with size-changeable slot was designed and manufactured in our lab for evaluating the plugging performance of the developed new diverter under flow conditions. The slot size can be changed from 0.5 mm up to 10 mm.

Results and Discussion

Diverter Properties

Size and Shape. In this work, a series of diverter particles have been synthesized under different conditions and using different mold to control the shape and size of the final product. Figure 2 shows the microscope image of the diverter particles with small size and the photo of diverter particles with larger size. The polymer particles generated in the synthesis solution are overall small and spherical, as observed via a

digital microscope under 20X magnification (Figure 2a). Through the microscope image, we have analyzed the particle size distribution of the synthesized diverter material by counting out about 100 particles. As shown in Figure 2b, the diverter has a narrow size distribution, mainly in the range of 0.25-1.5 mm. More precisely, the particle size in 0.25-0.5, 0.5-0.75, 0.75-1.0 and 1.0-1.25 accounts for 23%, 32%, 22% and 18%, respectively. When an engineered mold is used, we are able to better control the shape and size. As shown in Figure 2c, spherical balls with a diameter of 3.5, 5 or 8 mm can be easily obtained. The results indicate that the particle size of new diverter can be easily tuned and tailored to practical need. Furthermore, the developed method in preparation of new diverter is simple and easy to scale up, thus is promising for real applications.

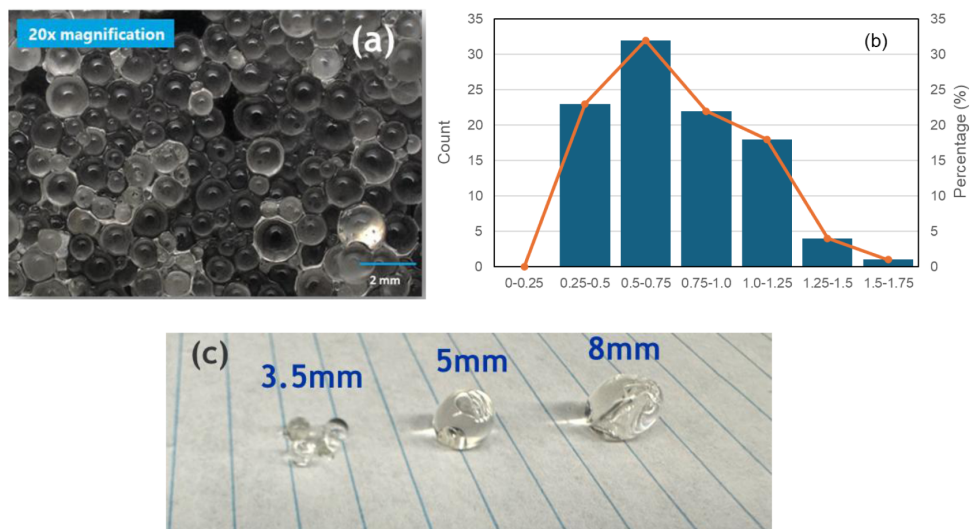


Figure 2—(a) Microscope image and (b) photo of the diverter particles synthesized in this work.

Mechanical Property. As above-mentioned, the elasticity of the newly synthesized material is critical for its use as an expandable diverter. Thus, we have conducted the compression test on the polymer particle using a universal test machine, Instron 5966. The obtained stress-strain curve is presented in Figure 3. In this test, the maximum compression stress was pre-fixed at 2000 psi, which means when the compression reaches 2000 psi, the piston stops and immediately returns to its original position at the set rate. As seen from the figure, the material can be easily compressed to about 15% without much stress implemented. Then a graduate increase in the strain change region of 15-23%. After that, a quick increase in the stress is observed when the strain is over 25% and quickly reaches to the pre-set compression stress limit, i.e., 2000 psi.

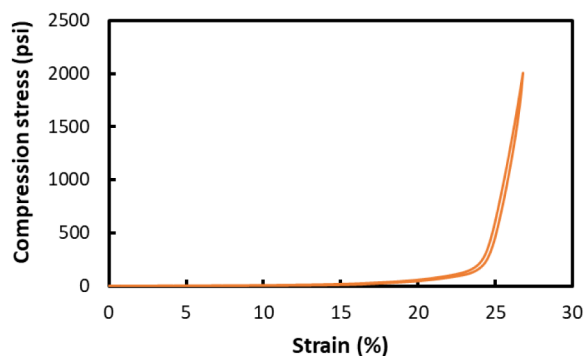


Figure 3—Stress-strain curve of the new diverter under compression.

The compression process was also recorded with a digital camera. The snap photos of the compression process are presented in Figure 4. Photos starting from the left to the right display the compression process from the beginning, reaching the maximum pre-set compression stress, going back and ending at the original Under the current compression conditions, the material shows good flexibility. After compression, it completely goes back to its original state, confirming the excellent elasticity of the diverter material.

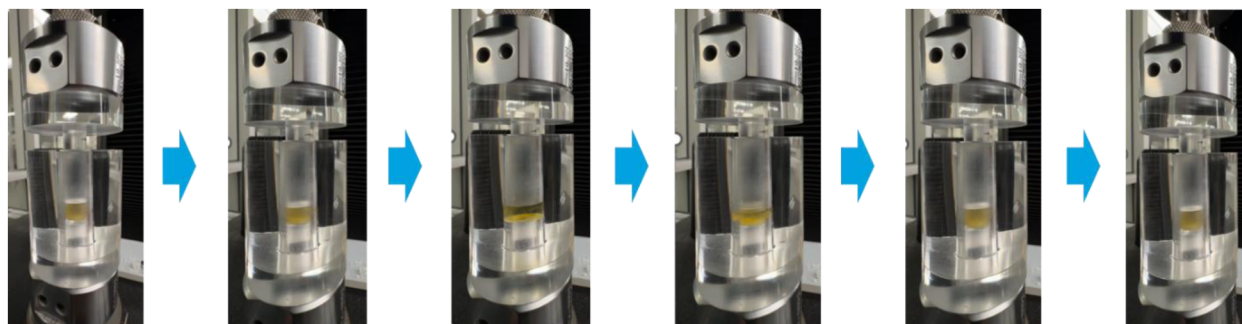


Figure 4—Snap photos of the compression process on the developed diverter.

According to the stress-strain curve, the energy loss coefficient ($\Delta U/U$) can be calculated, which is about 16%. Besides, another important parameter, Young's Modulus (E_Y) can be derived from the slope of the loading curve at the beginning stage, which is shown in Figure 5. The estimated E_Y is about 0.6 psi/%.

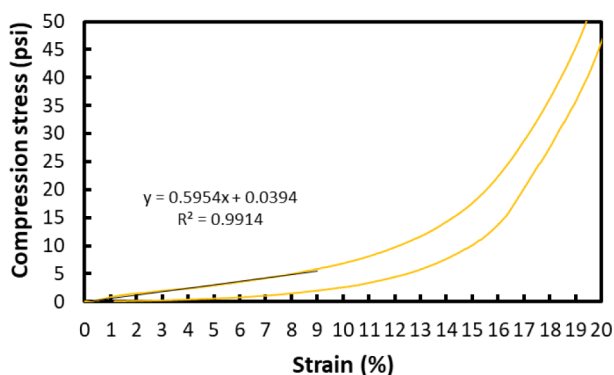


Figure 5—Estimation of the Young's Modulus for the developed diverter.

To examine the strength of the developed material, we further increased the maximum compression stress to 4000 psi and conducted 4 cycles of compression-release process. The results are shown in Figure 6. For comparison, a pack of 100-mesh sand was also examined using the same procedure, which is also presented in Figure 6. For the new expandable diverter, it can easily withstand the compression stress of 4000 psi. The stress-strain curve for the second cycle is almost overlapped with the first cycle. However, a slight strain change is observed in the third cycle ($\sim 2\%$) and in the fourth cycle (about another 0.5%). All four cycles have very similar shape, indicating the synthesized polymer material has excellent expandability and elasticity. On contrast, sand sample has very little expandability, the maximum strain of which is only about 1.5% . The deformation capability of the new diverter is 16 times larger than the sand sample tested in this work.

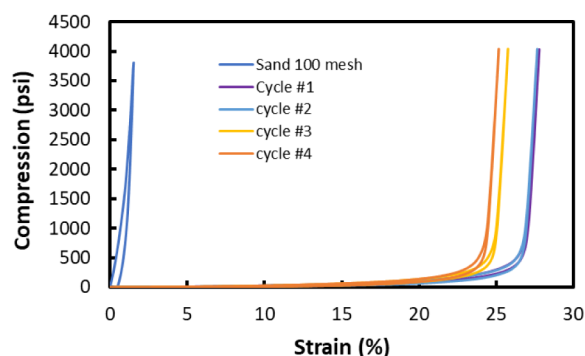


Figure 6—Stress-strain curves of the new diverter in 4 cycles under compression along with the data for 100-mesh sand.

The energy loss coefficient ($\Delta U/U$) of the new diverter for the four cycles is 15%, 22%, 20% and 21% for the 1st, 2nd, 3rd and 4th cycle, respectively. The value is constant, and much smaller than that of the sand sample, which is about 43%. The results indicate that the developed material maintains its high elasticity under compression stress of 4000 psi.

The Young's Modulus (E_Y) of the sand sample is estimated from the stress-strain curve. As shown in Figure 7, the estimated E_Y for the sand sample is about 1163.1 psi/%, which is about 2000 times higher than that of the new diverter. It confirms the excellent elasticity of the developed diverter.

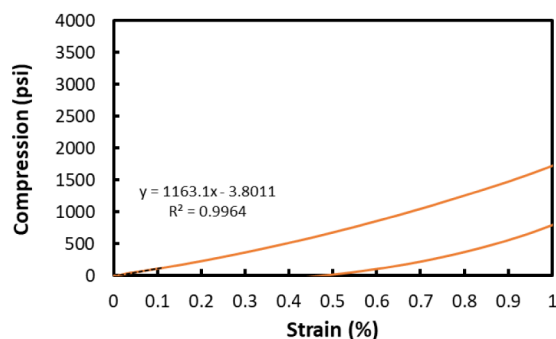


Figure 7—Estimation of the Young's Modulus for the 100-mesh sand sample.

FTIR Analysis. The chemical composition of the synthesized diverter was characterized by FTIR, which is shown in Figure 8. For comparison, the reactant was examined and is also plotted in the figure. Over the reactant, several distinct IR bands can be clearly observed. The IR band at 3275 cm^{-1} can be attributed to the stretching vibration of N-H bonding (Durand, D'Orlando et al. 2023). Two bands at 2978 and 2889 cm^{-1} are related to the C-H stretching vibrations in $-\text{CH}_2-$, CH_3 -groups (Himmelsbach, Khalili et al. 2002, Allan, Daly et al. 2019). The IR band at 2276 cm^{-1} can be attributed to the $\text{N}=\text{C}=\text{O}$ group, which is also reported at approximately 2270 cm^{-1} (Pitsevich, Shundalau et al. 2011, Durand, D'Orlando et al. 2023). It also has a strong C=O stretching vibration band at 1740 cm^{-1} and a strong C-O-C ether stretching vibration band at 1099 cm^{-1} (Allan, Daly et al. 2019, Durand, D'Orlando et al. 2023).

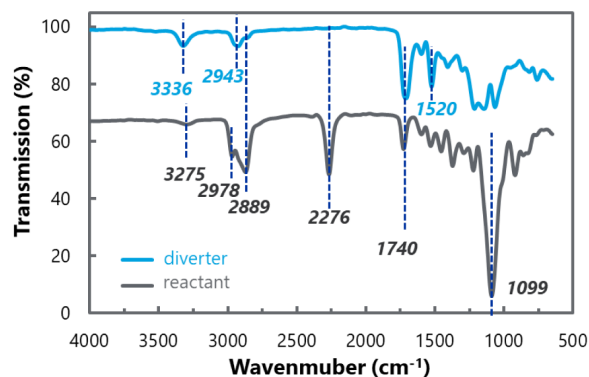


Figure 8—FTIR spectra of the developed diverter and the reactant E14 prepolymer.

After reaction, the most distinguished change in FTIR characteristic peaks of the diverter product is lack of the bands related to the $\text{N}=\text{C}=\text{O}$ group at 2276 cm^{-1} , which entirely disappeared from the diverter product. This peak is usually used as an indicator reflecting the completion of the reaction (Durand, D'Orlando et al. 2023). Here, such a change strongly suggests that the reactant has been completely consumed and converted into the final product. Another change is the band at 1099 cm^{-1} , which becomes much weaker compared to the reactant, suggesting the disappearance of the $\text{C}-\text{O}-\text{C}$ ether groups. The diverter has a strong IR band at 3336 cm^{-1} which can be attributed to $\text{N}-\text{H}$ stretching vibration. The bands at 2943 and 2889 cm^{-1} still exist, but relatively weaker, indicating more other bonds such as $\text{N}-\text{H}$ bonds are formed after reaction. In addition, another strong IR band at 1520 appears, which is attributed to a stretching vibration of the $\text{C}-\text{N}$ bonding (Niesiobędzka, Głowińska et al. 2021).

Thermal Property. The thermal property of the developed diverter was determined by a thermogravimetric analysis (TGA) method, which the change in the sample mass is tracked as a function of temperature. Figure 9 shows the TG and differential thermogravimetric (DTG) profiles of the diverter sample. As seen, the TG and DTG curves show that the decomposition displays a three-step-mass loss tendency: the first peak at about 210°C , the second peak at 325°C which is the largest and a should peak at about 350°C . The first peak can be attributed to the degradation of polymer moiety with small molecules and/or unreacted diisocyanate, while the second and third peaks can be related to the degradation of urethane groups with different molecular mass (Niesiobędzka, Głowińska et al. 2021). Besides the decomposition peaks, the decomposition starting temperature can also be obtained from the TG and DTG curves, which is about 130°C . The TGA data show that the developed diverter has good thermal stability, thus should have no problem for use at the bottom of the wellbore with high temperature for diversion.

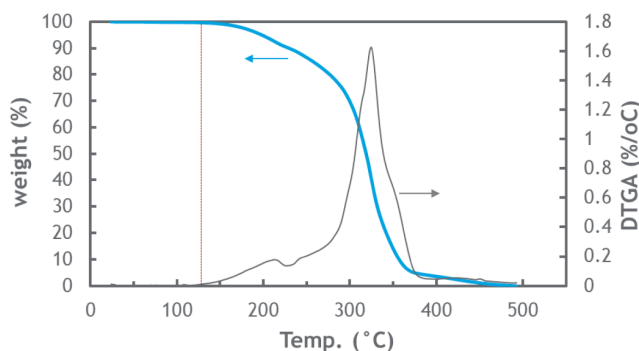


Figure 9—Thermogravimetric (TG) and differential thermogravimetric (DTG) profiles of the developed diverter.

Plugging Performance Evaluation

The developed diverter has been evaluated in the testing apparatus as illustrated in Figure 1. The output slots with 2 and 4 mm have been applied to examine the bridging and plugging performance. We observed over 500 psi increase in pressure during the plugging test for both 2 mm and 4 mm slots. The photo in Figure 10 clearly shows the plug of the slot with the diverter. It suggests that the developed diverter works well for the fractures with fixed widths.

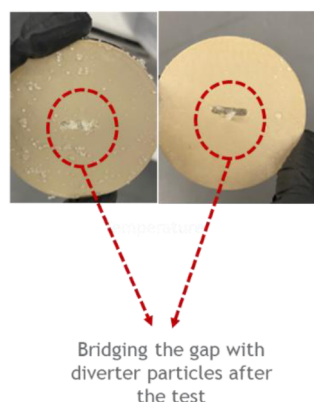


Figure 10—Photos of the plugged slot with the developed diverter after the plugging performance evaluation.

As outlined in the Introduction section, the key advantage of the developed diverter is that the diverter should work under a dynamic slot change condition. Thus, we further re-designed our testing apparatus so that the slot size can be tuned during the plugging performance test. Figure 11 shows the testing results. First, the diverter slurry was prepared by mixing diverter particles into the crosslinked guar as the fracturing fluid at the concentration of 2 ppg and filled into the testing stainless-steel tube. At the beginning, the slot size was controlled at about 0.3 mm. By flashing the fracturing fluid through the testing tube, a good sealing was achieved as observed the increased pressure to 700 psi for the first 10 minutes. After the pressure stabilized, the slot size was tuned and increased to 1.3 mm. The sealing pressure decreased a bit, but then quickly regained back to its set point of 700 psi. During the stabilization of 5 minutes, we did not observe any pressure drop, suggesting that the sealing is strong with the developed diverter.

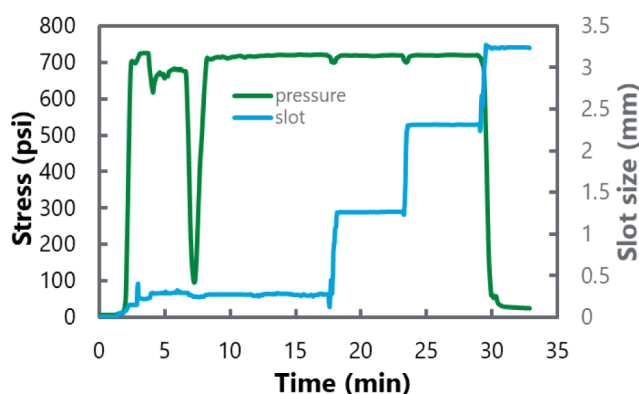


Figure 11—The stress and slot size as a function of time in plugging performance evaluation with the developed diverter.

Next, we continued increasing the slot size to 2.3 mm. Again, we just observed a very small disturbance in the sealing pressure with the increase of the slot size. After the slot size was set, the sealing pressure maintained its set point of 700 psi. During the stabilization of 5 minutes, we did not observe any pressure drop, suggesting that the sealing is strong with the developed diverter when the slot size increased to 2.3 mm.

However, when the slot size increased to 3.3 mm, the pressure was not able to be sustained. The pressure quickly decreased down to almost zero, as shown in Figure 11, indicating the diverter was no longer able to seal the slot. Additionally, we observed the diverter particles being pushed out of the slot and accumulated in the collection bottle.

It should be mentioned that the diverter tested for this study was a mixture of 25% particles with the size smaller than 2.8 mm and 75% particles with the size larger than 2.8 mm. It has been reported that the combination of diverter particle size could largely affect its plugging performance. Thus, as a continuation, we will conduct a more systematic study on the influence of particle size on the diversion performance of the developed diverter, which will be reported later.

Conclusions

In this work, we develop a new compressible/elastic/expandable diverter material to enhance the hydraulic fracturing efficiency by improving the diversion efficiency. By changing the synthesis conditions and reactant composition, we have successfully manufactured a series of the diverter materials with controllable shape and size ranging from 0.2 to several millimeters (mm).

The developed diverter has been characterized by various techniques for its physical and chemical properties including universal compression tests for the elasticity and mechanical strength, FTIR for the chemical composition and TGA for the thermal property. The characterization results confirm the excellent expandability of the developed diverter, which has polyurethane structure and has good thermal stability, suitable for high temperature wellbore.

The plugging performance of the developed diverter has been examined statically with fixed slot size and dynamically with tunable slot size. The material showed good plug/seal performance for the 2 mm and 4 mm slots. In the dynamic testing, it can successfully plug the slot opening to 3.3 mm. More systematic study on the influence of particle size on plugging performance will be conducted and reported.

The new diverter is a polyurethane based polymer, thus different type of degradation method for polyurethane polymer including hydrolytic, chemical, enzymatic, *in-vivo/in-vitro* oxidative, etc., (Xie, Zhang et al. 2019) would also be applicable to the diverter developed in this work. The degradation property will be studied systematically and reported.

In short, the newly developed diverter material shows great promise for diversion applications. Via its excellent expandability, it can improve the efficiency of multi-staged hydraulic fracturing, thereby reducing the total number of stages, and ultimately saving the associated operational time and costs.

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